Guglielmo Cerri

Senior Data Scientist · Generative AI, LLMs and Agents on Azure and Google Cloud

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PROFILE

Senior Data Scientist with over 5 years of experience across academic research, bioinformatics, financial services and IT consulting. I design and build production GenAI systems on Microsoft Azure and Google Cloud: RAG, document AI, multi-agent systems (A2A, MCP) and reusable AI libraries, often as tech lead of the delivery team. Background in machine learning, NLP and bioinformatics; open-source contributor and technical writer.

EXPERIENCE

Senior Data Scientist — Avvale Sep 2024 – Present · Remote

SAP and cloud consulting, digital transformation. Projects for clients in energy, insurance, certification, manufacturing and services.

- Led as tech lead the delivery, through go-live, of a multi-agent assistant for an energy major: a Master Agent that orchestrates Document and Data agents over A2A and answers geoscience questions with cited sources (15 of 16 in-scope test questions fully answered).
- Led as tech lead two GenAI platforms now in production on AKS: an AI knowledge base for an SAP AMS ticketing service (PostgreSQL/pgvector, LLM-based PII anonymisation) and an audit-review assistant built by reusing its components.
- Industrialised a GenAI document validator for an ESG questionnaire platform (Document AI OCR + Gemini on GKE), reaching 0.86 accuracy and 0.91 precision on a 611-case golden set, with a Vertex AI Pipelines evaluation framework.
- Designed GenAI microservices that automate claim investigation for an insurer (OCR, document classification at 86% accuracy on 844 documents, checks against policy terms, report generation) on Azure OpenAI.
- Created and maintain the central Python AI SDK of the Avvale AI practice (provider-agnostic LLM, embeddings, OCR and storage interfaces, LangGraph agents, model routing, cost tracking), used by the practice's AI microservices.
- Wrote the AI architecture and feasibility of GenAI and agentic offers in pre-sales; one multi-agent design was won and delivered.

Machine Learning Engineer | Bioinformatician — Cogentech Jun 2022 – Aug 2024 · Catania

Molecular diagnostics and oncology genomics lab (AIRC network).

- Built ML classifiers (mainly XGBoost) that detect tumour presence and origin from DNA methylation data, with a config-driven pipeline and signatures reduced from 23k to a few hundred regions; about 85% sensitivity at 94% specificity.
- Trained CNNs in PyTorch to tell DCIS and infiltrating carcinoma from normal ducts on whole-slide breast images (about 90% patch-level accuracy), with OpenSlide preprocessing and multi-centre stain normalisation.
- Built SynthSeqSimulator, a Python tool that inserts known mutations into synthetic NGS data to validate the lab's variant-calling pipelines; contributed 4 merged pull requests to the open-source BAMSurgeon tool.
- Built full-stack Streamlit web apps (MongoDB, Docker) that let lab staff launch and track Nextflow/nf-core pipelines, with about 30% less manual effort.
- Secondment at GENYO, Granada (Apr–May 2024), EU project diaRNAgnosis on liquid-biopsy biomarkers: containerised the project web app with Docker.

AI Engineer — Fire Group Sep 2021 – Jun 2022 · Messina

Italian credit-management group.

- Built a multi-modal document classifier for debt-collection documents (Tesseract OCR, FastText text model, EfficientNet image fallback, keyword rules) over 20 classes, served as a RabbitMQ worker in Docker.
- Trained a custom Italian NER model with spaCy on legal and payment documents and served it through a Flask API.
- Built collection-scoring models with LightGBM (AUC 0.908 on a utility portfolio) and Kedro pipelines for recovery-rate prediction.
- Developed a Rasa chatbot for customer support on Kubernetes, and PyQt5 tools to label PDF and image documents.

Bioinformatician | Research Fellow — University of Verona Nov 2020 – Aug 2021 · Remote

- Built an imaging genetics pipeline in R, Python and PLINK for Parkinson's disease (PPMI data): quality control, PCA, GWAS with logistic and linear regression models, and GSEA, linking genetic variants to brain imaging traits.

OPEN SOURCE AND WRITING

- **Khazad**: open-source semantic cache for LLM API calls on Redis 8 Vector Sets, with no change to application code (34 GitHub stars). **TubeTrim**: local YouTube summariser on open-source LLMs.
- Merged contributions to BAMSurgeon and featurewiz; issues and fixes reported in Google ADK and the Vertex AI SDK.
- 15 technical articles on Medium (Level Up Coding, Python in Plain English, AI in Plain English, Artificial Corner, MLearning.ai).

SKILLS

GenAI and agents: Azure OpenAI, Gemini, RAG, LangGraph, LangChain, Microsoft Agent Framework, Google ADK, A2A, MCP, prompt engineering, LLM evaluation, guardrails

Cloud and platform: Azure (AKS, Functions, Document Intelligence, AI Search), Google Cloud (Vertex AI, GKE, BigQuery, Document AI), Docker, Kubernetes, Helm, Terraform, CI/CD

ML and data: Python, PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, spaCy, SQL, PostgreSQL/pgvector, MongoDB, Redis, FastAPI, Streamlit

Bioinformatics: Nextflow/nf-core, GATK, BWA, samtools, PLINK, R

EDUCATION

MSc Medical Bioinformatics, University of Verona, 2018–2020, 110/110 cum laude

BSc Computer Science, University of Catania, 2015–2018, 110/110 cum laude

State licence as Information Engineer (2022)

LANGUAGES

Italian: native · English: professional working proficiency

CERTIFICATIONS

Google Cloud Professional Machine Learning Engineer (2025)

Microsoft Certified: Azure AI Engineer Associate (2025)

Deep Learning Specialization (2024) · Pretraining LLMs (2024)

PUBLICATIONS

Cerri G. et al., "Mining genetic, transcriptomic and imaging data in Parkinson's disease", IEEE ICHI 2021 (oral and poster); full version on bioRxiv, 2021.